Aura Workforce – Horizon Portal

Phase 3 Project Report – Tasks 11 to 15

Full Stack Developer Track | Date: 06 November 2026

1. Intern and Programme Details

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Project	Aura Workforce – Horizon Portal	**Report Period**	Phase 3 (Tasks 11–15)
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GitHub	github.com/NavaneethaSelvam	**Repository**	github.com/NavaneethaSelvam/horizon-portal

This report continues the previously submitted Phase 1 and Phase 2 reports and covers Phase 3 only (Tasks 11–15).

2. Project Overview

Aura Workforce – Horizon Portal is a role-based workforce management web application that supports employee operations, task management, payroll, finance, workload planning, and AI-assisted insights. Phase 3 extends the existing Horizon Portal with five operational modules, focusing on usability, data accuracy, role-based access, responsive design, and secure handling of workforce information.

3. Technologies Used

Frontend	Next.js 15, React 19, TypeScript, Tailwind CSS, dnd-kit (drag-and-drop)
Backend / Data	Node.js, REST API, Prisma ORM, SQLite

4. Phase 3 Summary of Tasks

Task	Module	Key Outcome	Access
11	Payroll Console	Payroll run overview and status; separate from employee payslip viewing	HR, Admin
12	Finance Console	Invoice overview, search and status filtering	Admin only
13	Team Task Board	Five-column Kanban with drag-and-drop, filters, optimistic updates and conflict handling	Role-based
14	Workload & Capacity	Workload, approved leave, sprint totals, capacity estimates and overload warnings	Role-based
15	AI Insights	Aggregate attrition-risk, anomaly and financial-risk insights with privacy suppression	HR, Admin

5. Task Details

Task 11: Payroll Console

- Developed a payroll management workspace for payroll operations.
- Added payroll run overview and status information.
- Restricted payroll access to authorized HR and Admin roles.
- Kept payroll management separate from employee payslip viewing.

Task 12: Finance Console

- Developed a finance workspace for invoice management.
- Added invoice overview, search, and status filtering.
- Implemented Admin-only access for finance operations.

Task 13: Team Task Board

- Developed a Kanban-style team task board with five columns: To Do, In Progress, Blocked, Done, and Cancelled.
- Implemented drag-and-drop task movement using dnd-kit.
- Added team and sprint filters.
- Implemented optimistic updates with rollback on failure, so the interface responds immediately and restores the previous state if a save fails.
- Added version-based conflict handling to protect concurrent updates.

Task 14: Workload & Capacity

- Developed a workload and capacity dashboard with team and sprint filters.
- Displayed employee workload and approved leave information.
- Added sprint workload totals and capacity estimates.
- Included overload warnings to help identify workload imbalance.

Task 15: AI Insights

- Developed an AI-assisted insights workspace for HR and Admin.
- Added department-level attrition-risk summaries.
- Added grouped anomaly insights and aggregate financial-risk information where available.
- Implemented small-group privacy suppression.
- Avoided exposing employee names, IDs, and individual prediction records in the default dashboard.
- Included clear notices that AI outputs are estimates for human review, not confirmed findings or automatic employment decisions.

6. Security and Privacy

- **Role-based access:** implemented for all new modules.
- **Restricted data:** sensitive payroll, finance, and AI insight information is limited to authorized roles.
- **Aggregate insights:** the AI module uses aggregate views and privacy safeguards, including small-group suppression.
- **Human review:** AI-generated results carry disclosures that they are estimates requiring human review.

7. Testing and Validation

Validation	Method / Result
Automated test suite	60 tests passed
TypeScript validation	`npx.cmd tsc --noEmit`
ESLint validation	`npm.cmd run lint`
Production build	`npm.cmd run build`
Git whitespace validation	`git diff --check`
Browser verification	New modules verified in the browser
Role-based visibility	Visibility checks performed for AI Insights
Mobile responsiveness	Checked at 360px width
Drag-and-drop	Manual verification on the Team Task Board

8. Screenshots

Screenshots of the five modules were captured and are maintained separately in the project folder `docs/phase3-screenshots/`:

Task	Screenshot file
Task 11	`task-11-payroll-console.png`
Task 12	`task-12-finance-console.png`
Task 13	`task-13-team-board.png`
Task 14	`task-14-workload-capacity.png`
Task 15	`task-15-ai-insights.png`

9. Conclusion

Phase 3 successfully extends the existing Aura Workforce – Horizon Portal with Payroll, Finance, Team Task Board, Workload & Capacity, and AI Insights modules. The implementation focuses on role-based access, reliable task updates, operational visibility, responsive user experience, and responsible use of AI-generated insights.

Source code: github.com/NavaneethaSelvam/horizon-portal